

Jaime Lopez Ruiz

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EDUCATION

Imperial College London

London, England

(MEng) Mechanical Engineering with Nuclear Engineering and a Year in Industry

Oct. 2022 – May 2027

- Expected 1st Class (+70%)
- Relevant Modules: Mathematics & Computing, Mechatronics, Dynamics & Vibrations, Thermo-fluids, Embedded C for Micro-controllers, Nuclear Chemical Engineering
- Programming project in C on PIC18 microcontrollers: developed autonomous maze-navigating buggy
- Nuclear Breeding Blanket Literature Report: Reviewed technologies for tritium breeding and heat extraction in fusion reactors, informed by discussions with lead researchers at the UK Atomic Energy Authority.
- Imperial College Formula Student Oct. 2023 – Jun. 2024
 - In chassis team, brakes division. Design in SolidWorks, manufacturing in workshop
- Imperial College Rocketry (3rd year project) Oct. 2024 – Jun. 2025
 - First student rocket team to hot fire vertically in UK history
 - Designed, built, and tested propellant feed system and body structure, integrating subsystems

Runnymede College

Madrid, Spain

British A-Level System & Spanish Baccalaureate

Oct. 2020 – Jun. 2022

- A Levels: A*A*AA; Spanish Baccalaureate: 13.318/14 (95%)

WORK EXPERIENCE

PerkinElmer

High Wycombe, England

Mechanical Design Engineer (13-month Industrial Placement)

Sep. 2025 – Present

- Currently developing and prototyping high-precision mechanical components/assemblies in SolidWorks for IR/UV spectroscopy instruments
- Collaborating with scientists and senior engineers on troubleshooting and verification of prototype instruments, integrating mechanical, optical, and electro-mechanical subsystems

Oxford University

Oxford, England

Summer Research Placement

Jun. 2025 – Aug. 2025

- Awarded UKAEA FOSTER grant for undergraduate nuclear engineers
- Investigated irradiation-induced damage of SiC in nuclear reactor conditions using Raman spectroscopy
- Produced research report aimed at peer-reviewed publication to inform future nuclear materials R&D

Visual 50 (visual50.com)

Madrid, Spain

Summer Internship

Jun. 2024 – Sept. 2024

- Developed Python program for IP network congestion detection, based on data from a network optimization product, using Machine Learning classification techniques and reinforced learning.
- Developed UNIX skills, as well as Python oriented towards Machine Learning.

EXTRACURRICULAR COURSES & CERTIFICATIONS

- Machine Learning with Python - From Linear Models to Deep Learning – MITx online course
- Modern Robotics: Mechanics, Planning, and Control Specialization – Northwestern online course
- Nuclear Energy: Science, Systems and Society – MITx online course

SKILLS

Technical

- Python, C, MATLAB, SolidWorks, FEA, HTML, CSS, Arduino

Languages

- Spanish (Native Fluency), English (Native Fluency), French (Fluent – DALF C1)